Run report: MutantXL closure degree against m·ℓ, up to m·ℓ = 30

Study `research/dreg_closure_slope_20261018/`, pre-registered in `PREREGISTRATION.md`. **Run** run-20261018T0618Z (2026-10-18 06:18 UTC; queue log 2026-10-18T06:17:56Z to 2026-10-18T13:29:43Z). **Commit** `9a44015c1`. **Class:** stage diagnostic, with no `S`, no rho ratio and no speed claim.

What ran

- **Draws.** `run.sh LADDER CLOSURE gen` generated the draws with seed 20261018 (exact solution count; XL built to degree 3 only), committed before any closure ran. Files: `draws/*.jsonl`.
- **Instrument check (P4).** `run.sh … check` replayed committed dense degree-4 refutations on the sparse engine: 1 of 2 `runs/check-*` records non-empty.
- **Measurement.** `run.sh … measure` ran one process per (draw, degree), as `prlimit --as=7000000000 timeout 10800 dreg_closure --sparse D --max-entries 350000000 --draw K`, degrees 4, 5, 6 in turn (5, 6 for the committed m·ℓ = 21 cells), each draw stopping at its first refutation. 18 of 21 `runs/*.D*.jsonl` measurement records are non-empty; 0 record a closure refutation and 18 hit the entry cap.
- **Control (P5).** `run.sh … control`: 0 of 0 `runs/control-*` records non-empty.
- **Cells.** The frozen table in `PREREGISTRATION.md`: committed anchors at m·ℓ = 12–18, fresh cells at 20, 24, 27, 30, and the two committed m·ℓ = 21 cells.
- **Host.** The 4-core, 15 GB container, which is not isolated (`AGENTS.md` §10). Wall times are practicality notes only.
- **Scorer.** `cargo run --release --example dreg_score -- slope research/dreg_closure_slope_20261018/runs/*.D*.jsonl`, output in `score-output.txt`.

Completion

Processes (from `runs/queue.log`): 16 sparse closure runs on the two committed m·ℓ = 21 cells, all exit 0 and all `capped` (degree 5 after 168–277 s, degree 6 after 102–121 s); 3 runs on fresh m3 (22, 8) draw 2 (degree 4 exit 0 capped after 8,466 s; degree 5 exit 0 capped after 82 s; degree 6 exit 134, "memory allocation failed" under the 7 GB address-space limit); draws 4 and 5 of that cell killed by hand when the queue was stopped after 155 and 13 CPU-minutes (empty records, unmeasured); draw 7 and every later cell not started. P4: check m3 (18, 6) draw 0 exit 0, refuted at degree 4 after 1,325 s (rank 64,433, 6,863 falls, 400,372 products); check m4 (13, 4) draw 1 killed by a worker restart after 80 minutes (empty record). Peak RSS was not recorded by the harness for sparse runs; `ps` during the runs showed 0.4–0.8 GB. Slowest finished draw: m3 (22, 8) draw 2 at degree 4, 8,466 s. No wall-clock claim.

Measured values

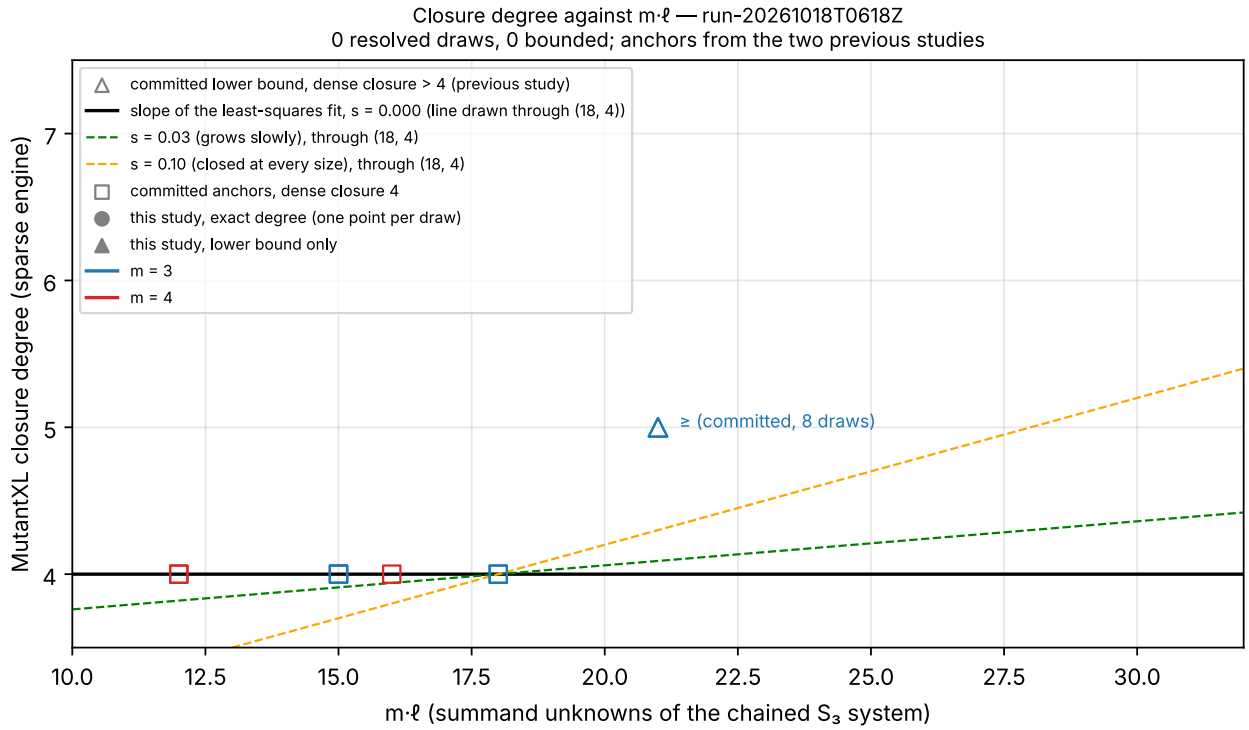


Figure 1: Recorded values from `runs/*.D*.jsonl`, run run-20261018T0618Z. Closure degree per draw against $m \cdot \ell$; squares are the committed anchors, circles exact degrees, triangles lower bounds. The solid line has the least-squares slope read from `dreg_score slope`, drawn through (18, 4) for display (not the fitted intercept); dashed lines are the two registered thresholds, drawn the same way. Source: `plot.py` (layout only).

Per-cell, per-draw degrees (measured)

m	(n, ℓ)	m·ℓ	N	degree per draw	cell	secs per draw
4	(12, 3)	12	36	4 4 4 4 (committed, dense)	resolved at 4	—
4	(14, 3)	12	40	4 4 4 4 (committed, dense)	resolved at 4	—
3	(11, 5)	15	26	4 4 4 4 (committed, dense)	resolved at 4	—
3	(13, 5)	15	28	4 4 4 4 (committed, dense)	resolved at 4	—
3	(15, 5)	15	30	4 4 4 4 (committed, dense)	resolved at 4	—
4	(13, 4)	16	42	4 4 4 4 (committed, dense)	resolved at 4	—
3	(16, 6)	18	34	4 4 4 4 (committed, dense)	resolved at 4	—
3	(18, 6)	18	36	4 4 4 4 (committed, dense)	resolved at 4	—
3	(19, 7)	21	40	unmeasured unmeasured unmeasured unmeasured	unmeasured	277 269 247 264
3	(21, 7)	21	42	unmeasured unmeasured unmeasured unmeasured	unmeasured	220 223 220 231
3	(22, 8)	24	46	unmeasured unmeasured unmeasured	unmeasured	8548 0 0

Registered readings (from `dreg_score slope`)

```
Slope s = 0.0000 degrees per summand unknown (m·ℓ), least squares over 8 cells (8 anchors at degree 4, 0 measured here)
Registered verdict: nothing resolved above the anchors: undecided
```

Prediction outcomes P1 to P5

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P1 (every resolved draw at m·ℓ = 20, 21 reads 5): 0 of 0 → unmeasured
P2 (a resolved cell with m·ℓ ≤ 30 reads 6): refuted on the resolved cells (or unmeasured)
P3 (s ≥ 0.10): unmeasured
P4 (sparse degree-4 replays of committed dense refutations refute): 1 of 1 measured (2 runs) → incomplete (capped or killed runs are unmeasured)
P5 (no satisfiable draw refuted): 0 control runs, 0 refuted, 0 unmeasured
```

Distance remaining (derived)

Derived (cost model, w = 1; extrapolated degree 4 + s·(m·ℓ − 18) at ℓ = ⌈n/m⌉, read from the fit over m·ℓ ≤ 30):

n	m	ℓ	$\log_2$ budget per solve	admissible D (chained N)	admissible D (N = m·ℓ)	extrapolated degree	gap, degrees (chained)
73	3	25	13.3	none	2	4.0	—
73	4	19	18.3	2	3	4.0	+2.0
73	5	15	19.0	2	3	4.0	+2.0
73	6	13	21.4	2	4	4.0	+2.0
109	3	37	19.6	2	3	4.0	+2.0
109	4	28	27.6	3	5	4.0	+1.0
109	5	22	29.3	3	5	4.0	+1.0
109	6	19	33.7	4	6	4.0	+0.0
127	3	43	22.7	3	3	4.0	+1.0
127	4	32	30.7	4	5	4.0	+0.0
127	5	26	36.4	4	6	4.0	+0.0
127	6	22	39.8	5	7	4.0	−1.0
131	3	44	22.8	3	3	4.0	+1.0
131	4	33	31.8	4	5	4.0	+0.0
131	5	27	38.4	5	7	4.0	−1.0
131	6	22	37.9	4	7	4.0	+0.0
163	3	55	29.4	4	4	4.0	+0.0
163	4	41	40.4	5	7	4.0	−1.0
163	5	33	47.1	6	8	4.0	−2.0
163	6	28	52.5	6	10	4.0	−2.0
233	3	78	40.2	5	6	4.0	−1.0
233	4	59	59.2	7	10	4.0	−3.0
233	5	47	67.9	8	12	4.0	−4.0
233	6	39	72.3	8	13	4.0	−4.0

This table is **derived** from the stated cost model and the fitted slope. It is not a measurement, and it is not extrapolated beyond m·ℓ = 30 as a law.

Control

Soundness, as registered: the first satisfiable draw of each fresh cell, replayed with `--satisfiable --sparse 4` and `--sparse 5`. Outcome is the P5 line above. Records: `runs/control-*.d*.D*.{jsonl,log}`.

No control was run. Every registered control (first satisfiable draw of a fresh cell) is at N ≥ 46, where the degree-4 sparse closure caps after more than two hours, so a control would be unmeasured by construction; the previous study's 345-draw control covers the dense engine and the sparse engine's agreement test covers the sparse one.

Raw records

- `runs/*.D*.jsonl`: one line per (draw, degree), with rank, falls, products, maximum row weight, seconds and the `capped` flag.
- `runs/check-*`: P4 instrument check. `runs/control-*`: P5 soundness control.
- `runs/*.log`: peak VmHWM per process; `runs/queue.log`; `draws/*.jsonl`.
- `score-output.txt` (native `dreg_score slope`); `plot.py` and the graph sources beside this report.
- The interpretation is in `../../RESULTS.md`.